

/ - division [Quotient]:

$$5 / 2 = 2 \text{ [int / int = int]}$$

$$5.0/2=2.500000$$

$$5/2.0=2.500000$$

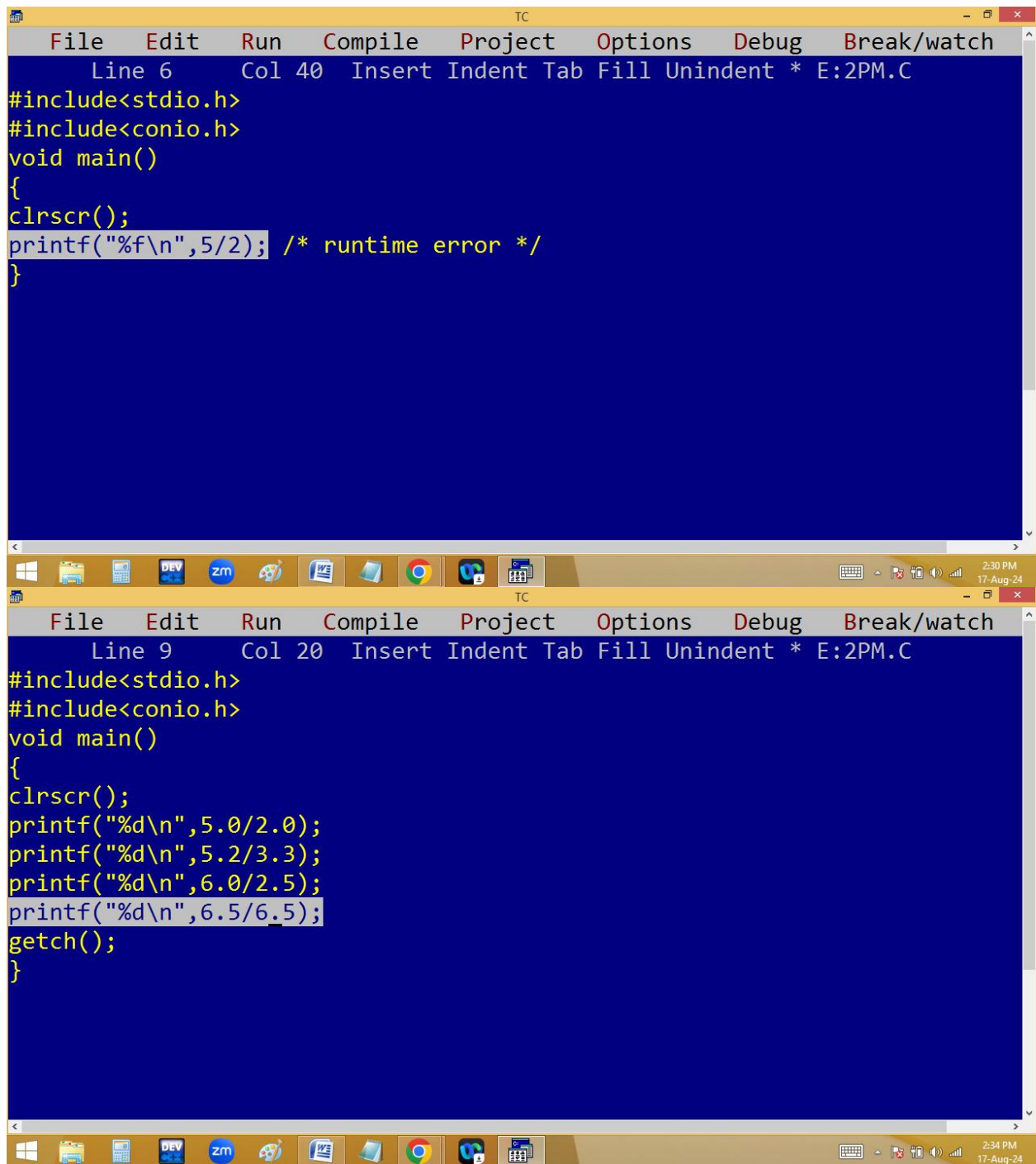
$$5.0/2.0=2.500000$$

Note: In C & C++ both operands are int then answer also int. if anyone or both are float the answer is float.

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code of a C program named E:2PM.C. The code includes `<stdio.h>` and `<conio.h>`, and defines a `main` function. Inside `main`, a float variable `a` is initialized to `3/2`. The program then performs several `printf` operations: it prints the integer part of `5/2` (which is 2), followed by the floating-point results of `5.0/2`, `5/2.0`, and `5.0/2.0` (all of which are 2.5). Finally, it prints the value of `a` using the format `"a=%f_"`, resulting in `a=1.000000_`. The bottom window shows the output of the program, which matches the `printf` statements in the code. The Windows taskbar at the bottom indicates the date is 17-Aug-24 and the time is 2:27 PM.

```
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float a=3/2;
clrscr();
printf("%d\n",5/2);
printf("%f\n",5.0/2);
printf("%f\n",5/2.0);
printf("%f\n",5.0/2.0);
printf("a=%f_",a);
}
```

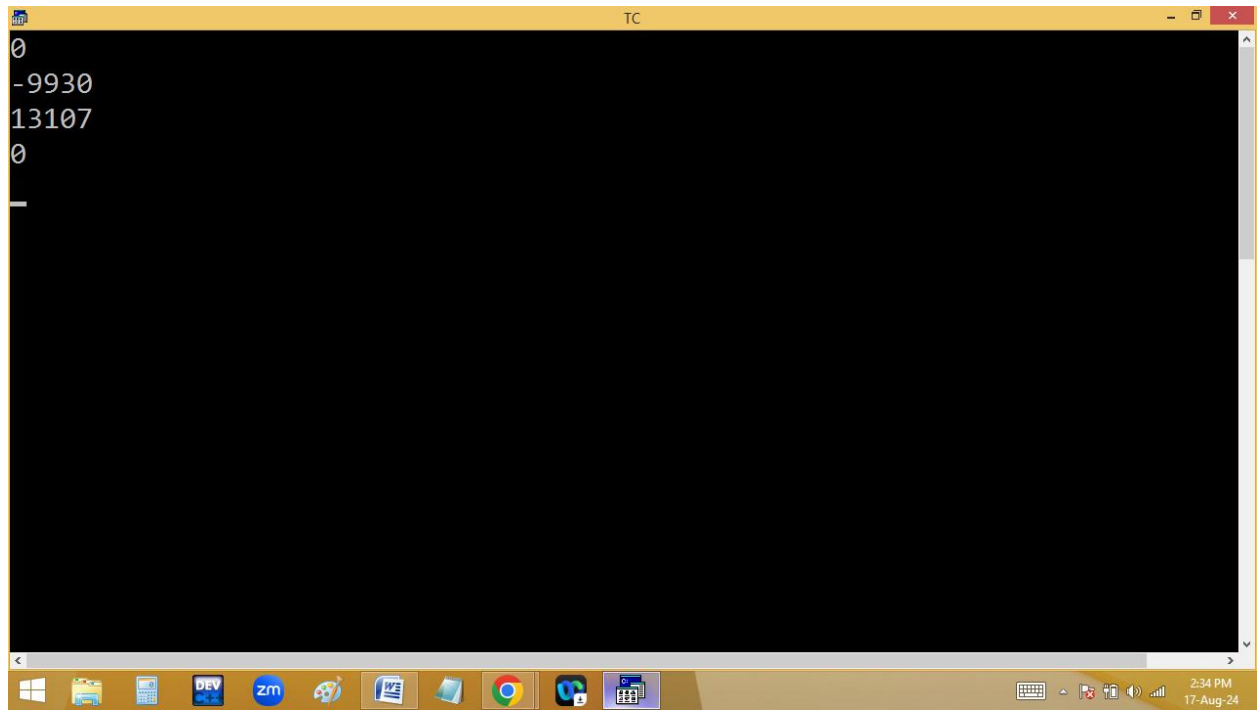
2
2.500000
2.500000
2.500000
a=1.000000_

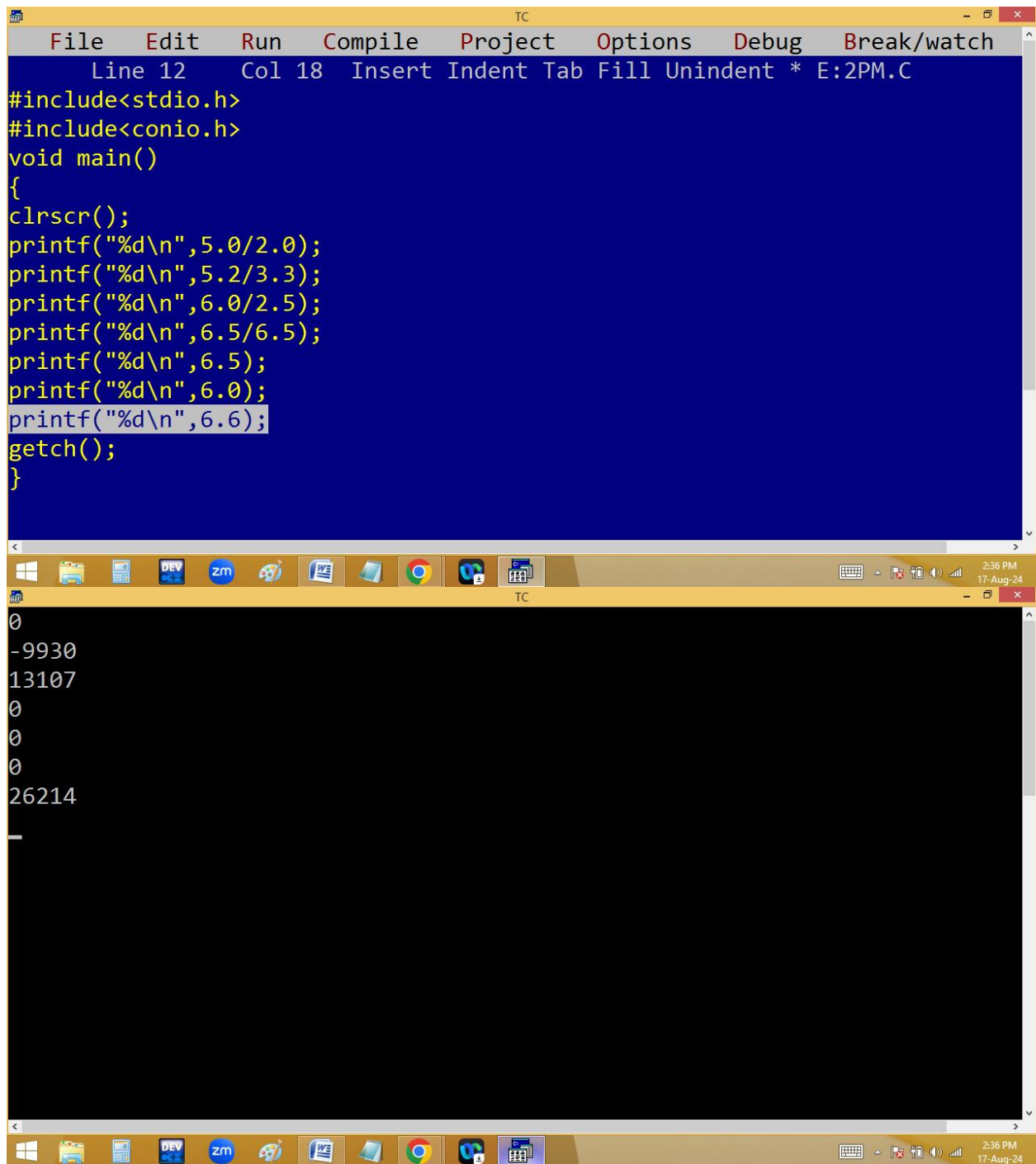


The image displays two instances of the Turbo C++ (TC) IDE. The top window, titled 'TC', shows a C program at line 6, column 40. The code includes `#include<stdio.h>` and `#include<conio.h>`, followed by a `void main()` function. Inside the function, `clrscr();` is called, and a `printf` statement is being edited: `printf("%f\n",5/2); /* runtime error */`. The bottom window, also titled 'TC', shows the same program at line 9, column 20. The code includes `#include<stdio.h>` and `#include<conio.h>`, followed by a `void main()` function. Inside the function, `clrscr();` is called, and four `printf` statements are present, with the last one being edited: `printf("%d\n",6.5/6.5);`. Below the `printf` statements, `getch();` is called, and the function ends with a closing brace. The Windows taskbar at the bottom of each window shows the time as 2:30 PM and 2:34 PM on 17-Aug-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 6 Col 40 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5/2); /* runtime error */
}
```

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 20 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5.0/2.0);
printf("%d\n",5.2/3.3);
printf("%d\n",6.0/2.5);
printf("%d\n",6.5/6.5);
getch();
}
```





The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5.0/2.0);
printf("%d\n",5.2/3.3);
printf("%d\n",6.0/2.5);
printf("%d\n",6.5/6.5);
printf("%d\n",6.5);
printf("%d\n",6.0);
printf("%d\n",6.6);
getch();
}
```

The bottom window shows the output of the program:

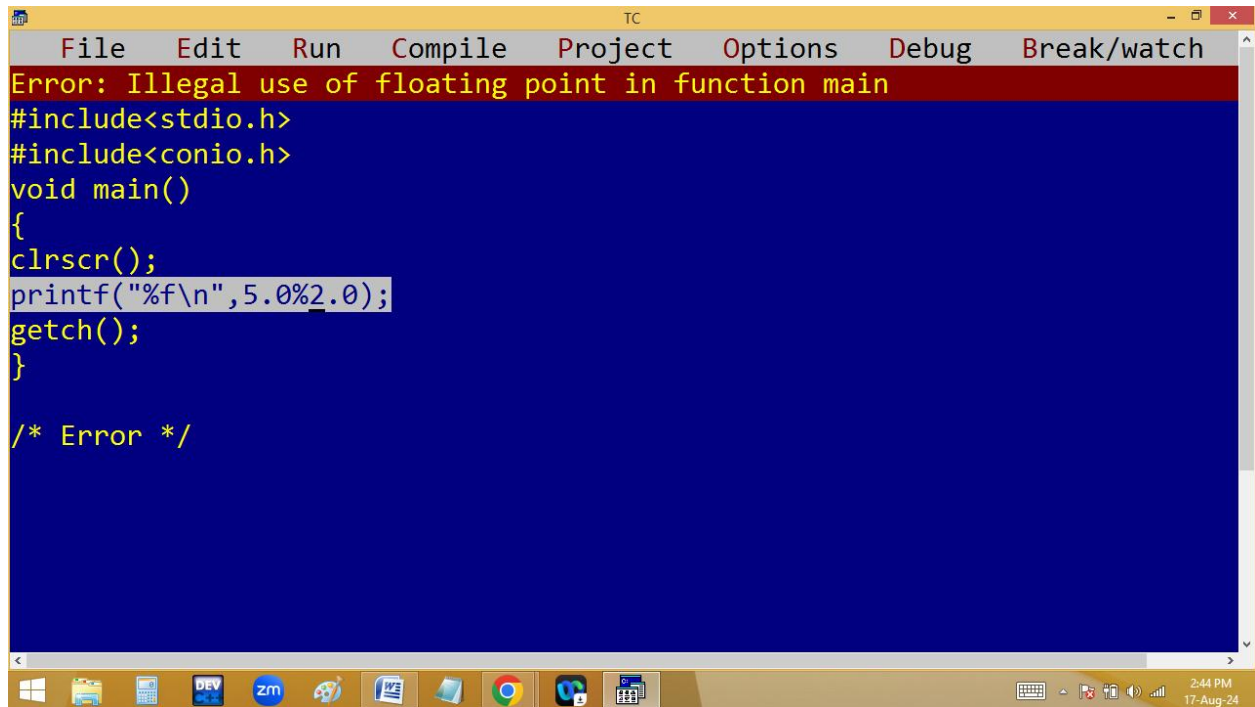
```
0
-9930
13107
0
0
0
26214
_
```

The Windows taskbar at the bottom indicates the time is 2:36 PM on 17-Aug-24. The taskbar includes icons for various applications like DEV, zm, and a calendar.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float    a = 3; /* implicit type casting */
int      b = 1.2; /* implicit type casting */
clrscr();
printf("%f\n", (float)5/2); /* explicit type casting */
printf("%f\n", 5/(float)2); /* explicit type casting */
printf("%f\n", (float)(5/2)); /* explicit type casting */
printf("%d\n", (int)5.0/2); /* explicit type casting */
printf("%f\n", a);
printf("%f\n", a);
getch();
}

2.500000
2.500000
2.000000
2
3.000000
3.000000

```



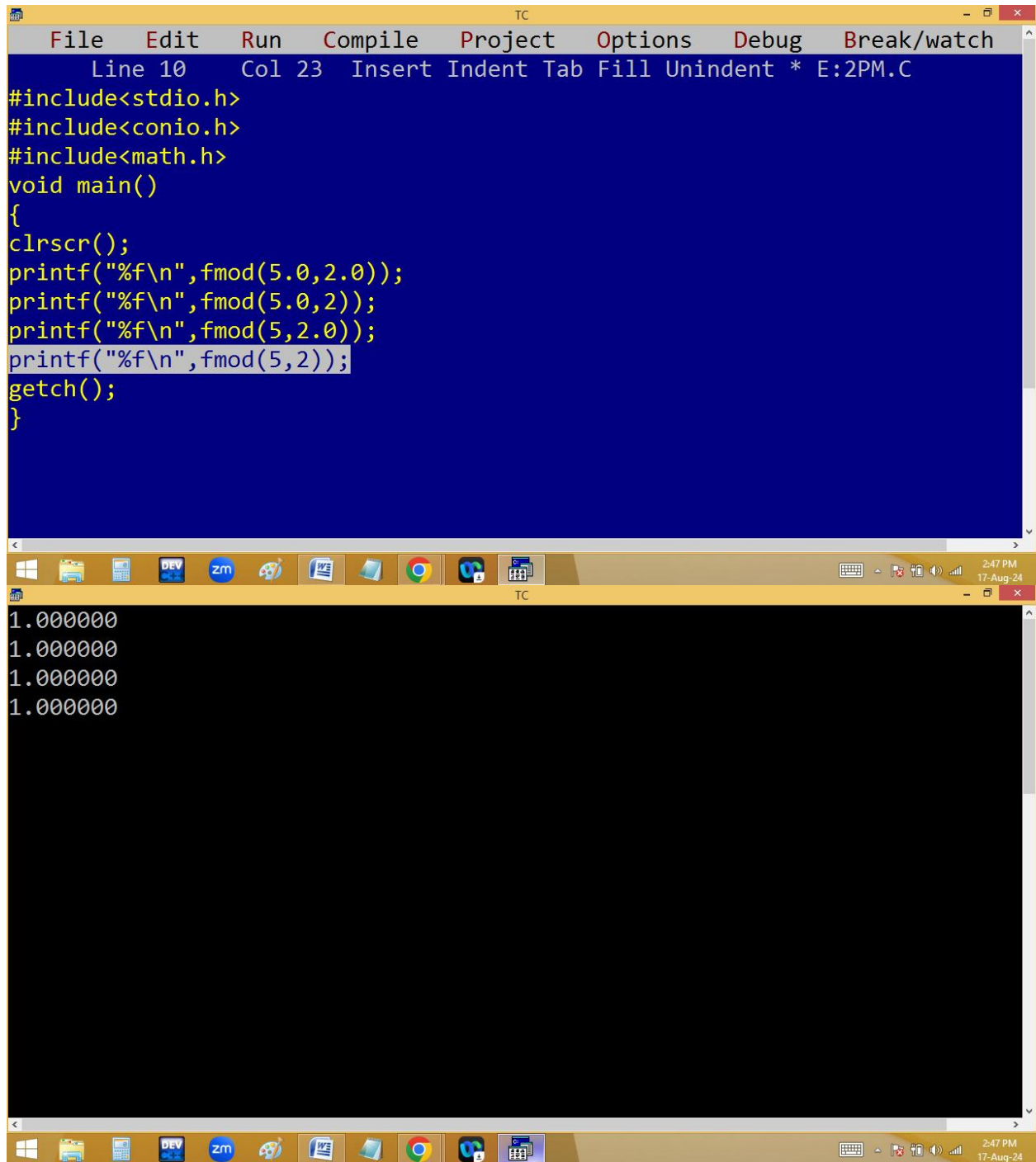
The screenshot shows the Turbo C++ (TC) compiler interface. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. A red error message banner at the top reads "Error: Illegal use of floating point in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5.0%2.0);
getch();
}

/* Error */
```

The Windows taskbar at the bottom shows various application icons and the system clock indicating 2:44 PM on 17-Aug-24.

Note: in c & c++ we can't do floating modules with % operator. For this we have to use fmod() available in <math.h>



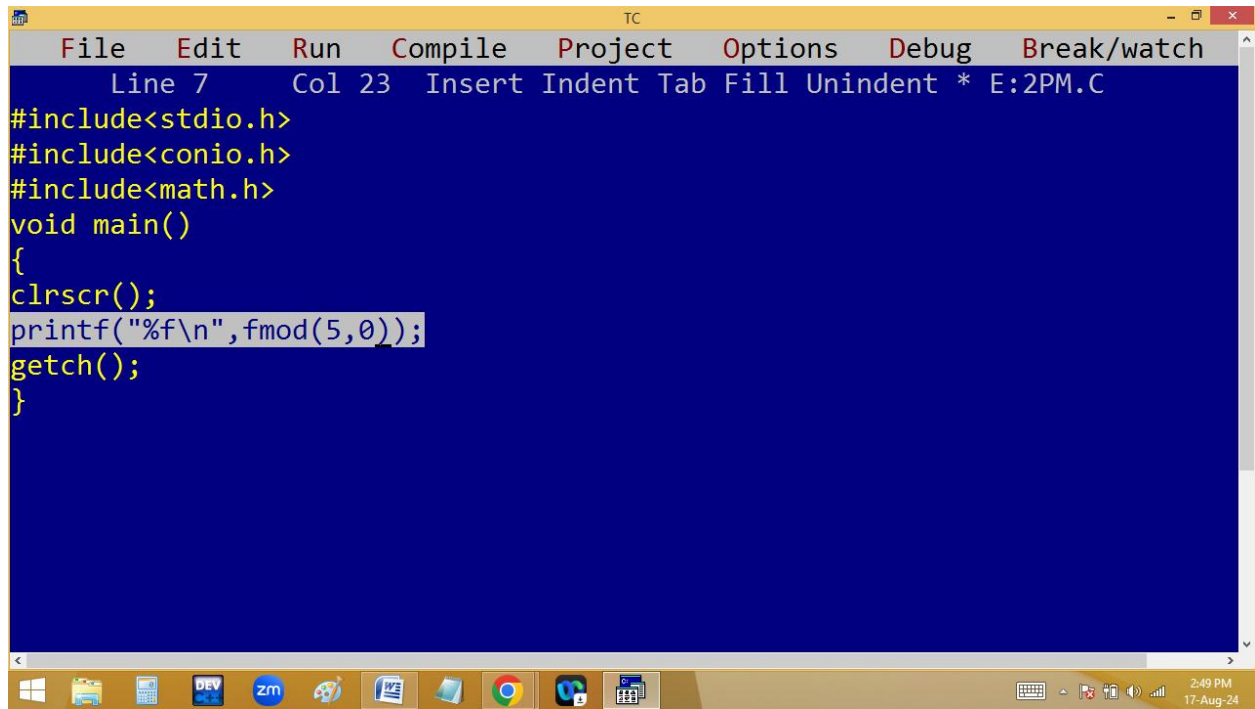
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5.0,2.0));
printf("%f\n",fmod(5.0,2));
printf("%f\n",fmod(5,2.0));
printf("%f\n",fmod(5,2));
getch();
}
```

The bottom window shows the output of the program, which consists of four lines of floating-point numbers:

```
1.000000
1.000000
1.000000
1.000000
```

The Windows taskbar at the bottom indicates the time is 2:47 PM on 17-Aug-24. The taskbar includes icons for various applications such as File Explorer, DEV, zm, and Google Chrome.



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
clrscr();
printf("%f\n",fmod(5,0));
getch();
}
```

2:49 PM
17-Aug-24

The screenshot shows the Turbo C++ (TC) IDE interface. The top window displays the output '0.000000'. The bottom window shows the source code and a compiler error. The error message is 'Error: Division by zero in function main'. The code includes `#include<stdio.h>`, `#include<conio.h>`, and a `main` function that calls `clrscr()`, `printf("%d",5%0);`, and `getch()`. The `printf` statement is highlighted. The Windows taskbar at the bottom shows the time as 2:49 PM on 17-Aug-24.

```
0.000000
```

File Edit Run Compile Project Options Debug Break/watch

Error: Division by zero in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d",5%0);
getch();
}
```

2:49 PM 17-Aug-24

$$123/10=12$$

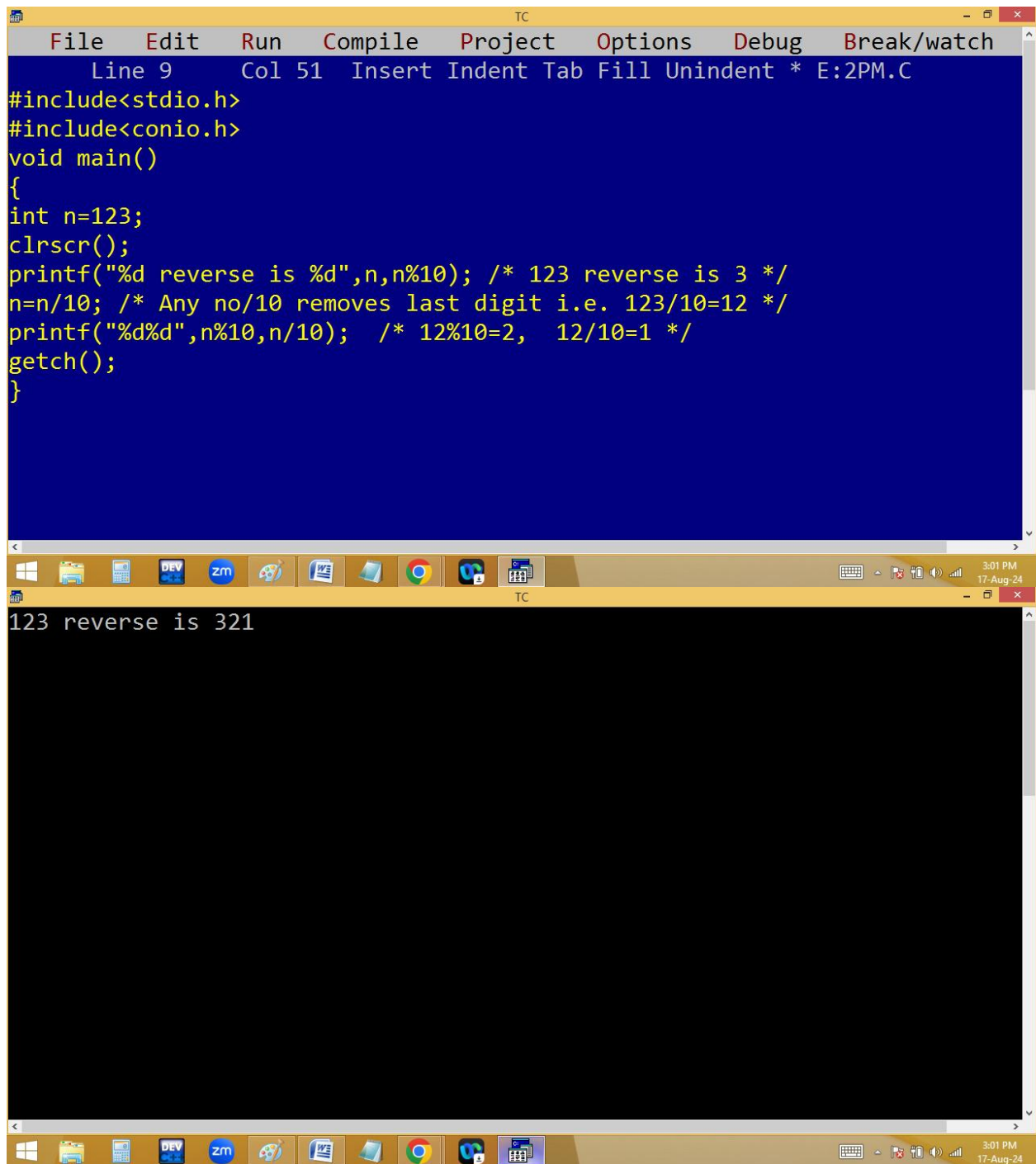
$$12/10=1$$

$$1/10=0$$

Note: Any no/10 removes last digit.

Write a c program to print a 3 digit no in reverse order without using loop:

Eg: 123 reverse is 321



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n=123;
clrscr();
printf("%d reverse is %d",n,n%10); /* 123 reverse is 3 */
n=n/10; /* Any no/10 removes last digit i.e. 123/10=12 */
printf("%d%d",n%10,n/10); /* 12%10=2, 12/10=1 */
getch();
}

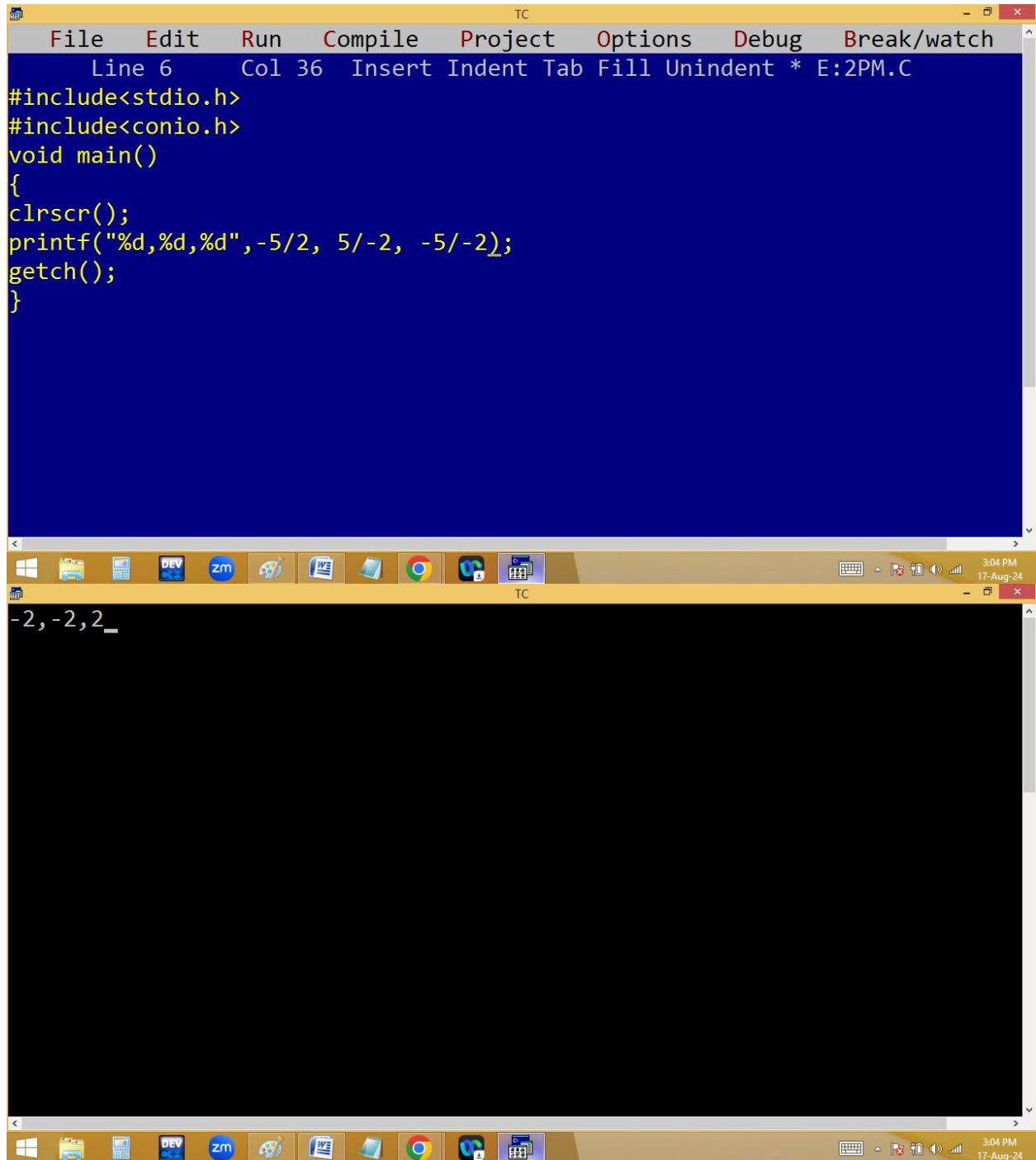
TC
123 reverse is 321
```

$$-5/2 = -2$$

$$5/-2 = -2$$

$$-5/-2 = 2$$

Note: In division any one operand is negative then result also negative. If both are negative result is positive.



The screenshot displays the Turbo C++ (TC) IDE interface. The top window, titled 'E:2PM.C', contains the following C code:

```
Line 6 Col 36 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d,%d,%d",-5/2, 5/-2, -5/-2);
getch();
}
```

The bottom window shows the output of the program: `-2,-2,2_`. The Windows taskbar at the bottom indicates the time is 3:04 PM on 17-Aug-24.

Relational operators [== (comparison), <, >, <=, >=, != (not equal)]:

They are used to check the given condition / expression is true or false. If condition is true always it return 1 and false it return 0.

```
TC
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",5==5);
printf("%d\n",5 == -5);
printf("%d\n",5.5 == 5.5);
printf("%d\n",5 == 5.0);
printf("%d\n",'5' == 5);
printf("%d\n","a" == "a");
printf("%d\n",'a' == 97);
printf("%d\n",'a'-32 == 'A');    /* 65==65 */
printf("%d\n",5==5==5);
printf("%d\n",5==5==1);
getch();
}
```

```
TC
1
0
1
1
0
0
1
1
0
1
1
```

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the source code editor, titled "TC", with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 10, Col 21, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",'a'>'A'); /* 97 > 65 */
printf("%d\n",'b'<='b');
printf("%d\n",'0'>=0); /* 48 >= 0 */
printf("%d\n",9 != 9);
getch();
}
```

The bottom window is the output console, also titled "TC", which displays the results of the program's execution:

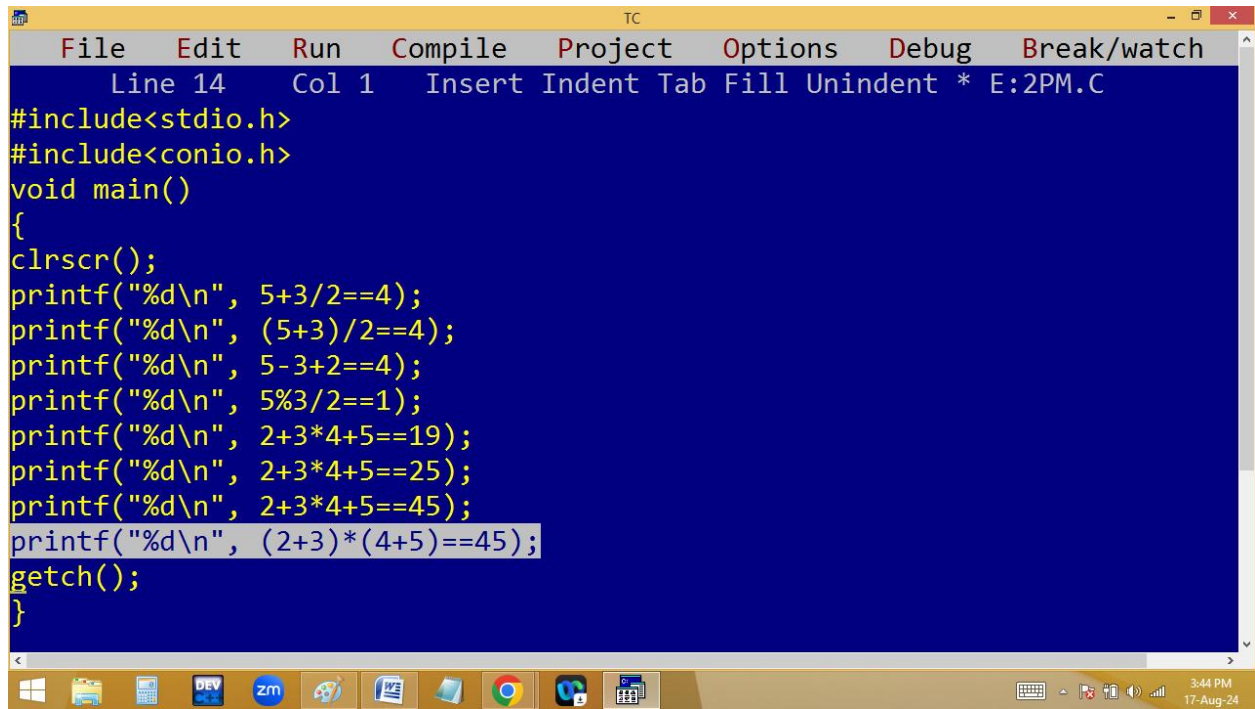
```
1
1
1
0
```

The Windows taskbar at the bottom of the screen shows the time as 3:23 PM on 17-Aug-24, along with various system icons and application icons in the taskbar.

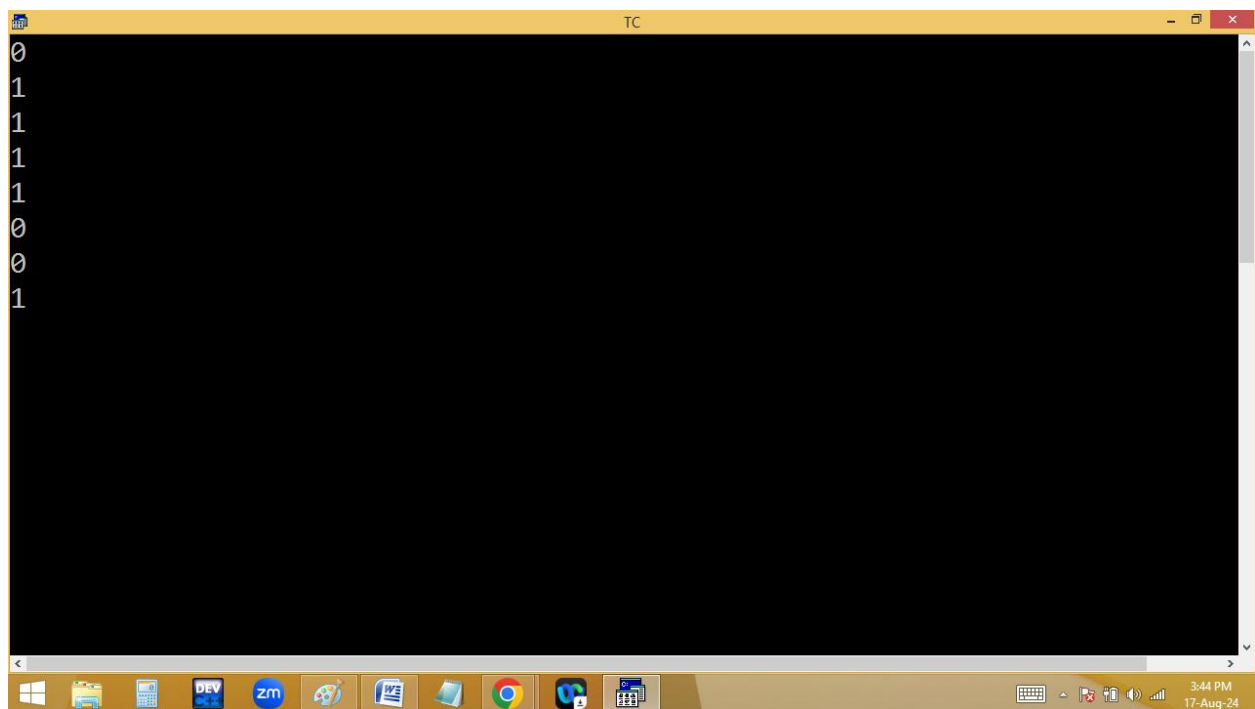
Operator precedence / Operator priority

(ASSOCIATION OF OPERATORS)

1. ()
2. +, -, ! (sign operators, unary operators)
3. ++, -- (pre increment & decrement)
4. *, /, %
5. +, - (Binary)
6. ==, !=
7. &&
8. ||
9. ?: (ternary operator)
10. =
11. ++, -- (Post increment & decrement)
12. , (comma)



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\\n", 5+3/2==4);
printf("%d\\n", (5+3)/2==4);
printf("%d\\n", 5-3+2==4);
printf("%d\\n", 5%3/2==1);
printf("%d\\n", 2+3*4+5==19);
printf("%d\\n", 2+3*4+5==25);
printf("%d\\n", 2+3*4+5==45);
printf("%d\\n", (2+3)*(4+5)==45);
getch();
}
```



```
TC
0
1
1
1
1
0
0
1
```

